MIL-V-364 57(DSA-DM)
4 January 1966

MILITARY SPECIFICATION

VACUUM PUMP, ROTARY, POWER DRIVEN, SMALL

This limited coordination Military Specification has been prepared by Defense Personnel Support Center, Directorate of Medical Materiel, 2800 South 20th Street, Philadelphia, based upon current available technical information, but it has not been approved for promulgation as a coordinated Military Specification. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of vacuum pump suitable for use in reducing pressure to 0.0003 millimeters of mercury.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproof, Kraft
PPP-B-601	Boxes, Wood, Wirebound
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

FSC 4310

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARD ASSOCIATION, INC.

ASA C73.11-1963 American Standard Dimensions Caps and Receptacles, 2 Pole, 3 Wire, Grounding Type, General Purpose.

(Application for copies should be addressed to the American Standards Association, Incorporated, 10 East Fortieth Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as herein specified.

3.2 Construction. The unit shall consist of a pump, motor and belt guard which shall be securely mounted on a common iron base. Power shall be transferred from the motor to the pump by means of a rubber V belt and pulley wheels. Belt and pulleys shall be aligned.

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3.2.1 Belt guard. Belt guard shall be fabricated of metal and shall ver the outside and top perimeters of the V belt and pulley wheels. Guard all be so constructed and mounted that it does not interfere with the eration of the unit.

3.2.2 Pump. Pump shall be housed in a metal case. Case shall have an take nipple which shall serve as the connection to the equipment to be acuated. Nipple shall be tapered with the distal diameter as the smallest ameter. Distal diameter shall be 3/8 ± 1/16 inch. Pump shall have a rmal operating speed of 350 revolutions per minute and a free air displace- nt of at least 10 liters per minute. Internal atmosphere and evacuated eas of the pump shall be separated by a reservoir of oil. Oil shall not in pump, but a sufficient amount shall be furnished for initial filling.

3.2.2.1 Oil. Flash point of oil shall be no lower than 450° Fahrenheit (F). por pressures shall be as follows: 140°F - 0.002 millimeters (mm) of rcury (Hg) (maximum), 180°F - 0.010 mm of Hg (maximum).

3.2.3 Motor. Motor shall be housed in a metal case, shall produce at ast 1/4 horsepower and shall be capable of operating on 110 volt, 60 cycle, ternating current.

3.2.3.1 Cord. An electrical cord shall be attached to motor housing. rd shall have an integral switch and shall be three wire, fitted with a ounding plug which is in accordance with ASA C 73.11-1963.

3.3 Performance.

3.3.1 Reduction of pressure. The unit shall reduce the pressure to 0003 millimeters (mm) of mercury or less when tested as specified in 4.4.1.

3.4 Dimensions. No overall base dimension shall exceed 18-1/2 inches be less than 9-1/2 inches. Overall height shall not exceed 13 inches or be ss than 10-1/2 inches.

3.5 Finish. All metal surfaces other than wheels, screws and other xtures shall be finished in blue gray enamel.

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3.6 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.7 Identification marking. Each unit shall have a nameplate permanently affixed to the unit on which the letters U. S., the name or registered trademark of the manufacturer, the contract or purchase order number, the type, serial number and rated characteristics; shall be permanently stamped.

3.8 Workmanship. Workmanship shall be first class throughout. Unit shall be free from defects which detract from its appearance or may impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that the supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examination and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Military Standard MIL-STD-105.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Pump	Vacuum	3.2.1	4.4.1	*	**

* Each pump shall be tested.
** Acceptance number shall be zero.

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4.3 Examination. The vacuum pumps shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Component part missing.
102 Component part not secured.
103 Oil supply missing.
104 Finish does not adhere; not free of blistering, checking, and peeling.
105 Unit not free of sharp edges.
106 Service data not legible.

Minor

201 Finish not free of scratches and bare spots.
202 Identification marking not present, complete, correct, legible, or permanent.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The vacuum pumps shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Vacuum. Previous to testing, the unit shall be subjected to break-in instructions outlined in the contractors service data. After the pump has been connected to a 25 liter container by means of the intake nipple and rubber tubing, the motor shall be turned on and allowed to run for ten minutes. Pressure in the container shall be determined by a pressure gauge which is calibrated and readable at a pressure of .0003 millimeters of mercury.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) vacuum pump, as specified, constitutes e unit of issue.

5.1.2 Level A.

5.1.2.1 Unit preparation. Belt shall be removed and wrapped in kraft per. As an alternate, belt and flywheel shall be immobilized by wiring main housing, in such a manner as to prevent unit from operating. A tag label painted in red shall be affixed to the wiring with a statement rning that the unit shall not be operated before applicable safety ecautions are observed and accomplished. All exterior finished surfaces all be protected by wrapping with kraft paper and cushioning with excelsior ds or flexible corrugated fiberboard or other suitable materials. Wrapping d cushioning material shall be secured by tying with cotton cord or tape, by sealing with kraft gummed paper tape. Work shall be accomplished in ch a manner as to insure against marring, scratching, or damage to the unit.

5.1.3 Level C. Units shall be packaged in standard commercial containers the size and kind commonly used, which will afford the degree of protection quired for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A. Each unit shall be overpacked in an exterior container signed for type 3 load, and constructed in accordance with Federal Specification P-B-601, table II, using type III, class 1 plywood, or PPP-B-621, table IV. osure and strapping shall comply with appendix of applicable specification. ch box shall be lined with a waterproof barrier fabricated of material nforming to the requirements of Federal Specification UU-P-271 and sealed th a suitable waterproof adhesive. The unit shall be adequately secured thin the exterior container, by means of bolts, wood braces, blocks, or her suitable devices, to insure against shifting and breakage, or marring surfaces.

5.2.2 Level B. Each unit shall be overpacked in an exterior container nstructed in accordance with Federal Specification PPP-B-601, table I, using pe III, class 1 plywood, or shall be designed for type 3 load and constructed accordance with Federal Specification PPP-B-621, table II. The unit shall adequately secured within the exterior container, by means of bolts, od braces, blocks, or other suitable devices, to insure against shifting, d breakage, or marring of surfaces.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.2 This specification covers the requirements for FSN 4310-438-4100.

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: N

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